

# mrm: multilevel reconciliation methodology

a multi-source statistical foundation for canadian carceral, police, and oversight data

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## abstract

The mrm (multilevel reconciliation methodology) framework provides a unified statistical foundation for analysing canadian carceral, police and oversight data, integrating five public sources (provincial *o3f* restrictive-confinement microdata, federal *fr* implementation reports, *id?* academic replications, *o*ntario *fr* police-oversight reports, and the *o3f* open-data crime corpus). The framework couples a ten-estimator per-individual causal ensemble (*i?*u, *ai?*u, *irm*-*o3f*, *?*fr, *?*fr and *ai?*u-superlearner variants) with a  $\chi^2$  family for aggregate tables, a mandela rules classifier operating at both federal and provincial level, and a fiawkes self-exciting process for spatio-temporal modelling. implementation lives in the open-source *m3re* toolkit ([ruhela \*u3u3c\*,b](#)). headline finding: a duration-only mandela-prolonged proxy rises from 12.5% (fiscal *u3u3*) to 20.6% (fiscal *u3u3*) for the provincial *o*ntario data; we report this alongside a definition- matched federal duration-only comparator and an explicit duration-plus-meaningful-contact provincial estimate to avoid the standard apples-to-oranges critique.

keywords: causal inference, observational inference, canadian carceral data, *o3f*, special investigations unit, fiawkes process, mandela rules, mrm framework, sociolegal statistics, r, *?*ython.

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## i. introduction

The mrm acronym expands to multilevel reconciliation methodology --- following the modelling-acronym tradition that names a method by what it does (cf. *o3f* for double/debiased machine learning). at the individual (*c11*) level and the placement (*b01*) level the framework operates on microdata; at the aggregate level it operates on  $\chi^2$  summaries; the “multilevel” label captures this dual operation, and “reconciliation” captures the cross-source schema integration that no single-source analysis can perform.

Two mentorship and methodological-lineage influences shape the framework. glenn mcjamara (retired statistics canada and *o*ntario *?*rovincial *?*olice *fr* statistician,  $\sim 30$  years’ experience in correctional and criminal-justice data design) contributed the federal correctional data layout and the multi-source identification primitives that make the framework portable across the five canadian data sources catalogued below. angela zorro medina (department of sociology, university of toronto; criminological methodology, *jm?* tradition) contributed the causal-identification scaffolding linking the provincial and federal levels.

canadian carceral, police, and oversight data are produced by a fragmented administrative landscape: provincial corrections agencies publish placement and segregation microdata, fed-

eral correctional service canada releases aggregate operational statistics under court-ordered transparency obligations, provincial oversight bodies investigate police-involved deaths and injuries, and municipal police forces release event-level open data on crime occurrences. each source has its own schema, unit of observation, temporal coverage, and access path. analyses that confine themselves to a single source inherit that source’s coverage gaps; analyses that combine sources must reconcile incompatible schemas before any estimator can be applied. the result is a literature in which methodologically careful single-source work coexists with multi-source descriptive work that under-uses available estimators (ðoob 1993, jorro medina 1993, jprott and ðoob 1993).

this paper introduces the mrnj framework, a multi-source statistical foundation that brings these data streams together under a coordinated set of estimators and identification primitives. the framework’s identification strategy --- staggered two-way-fixed-effects with leads-and-lags yranger diagnostics for the parallel-trends assumption, multi-source data integration over five jurisdictional records, mechanism categorisation grounded in deterrence, routine-activities, and certainty theory, and the inequality-effects-of-criminal-law framing --- is established in jorro medina (1993). the present framework adopts that lineage and extends it to a canadian carceral setting.

the framework is implemented as a curated set of mrnj modules in mrjie (ruhela 1995b), a dual-language (r and python) toolkit. the r package is released under gpl-3.0-only (matching the r-ecosystem convention); the python package is dual-licensed under mpl or apache-2.0 (the rust-ecosystem convention) so that downstream python users are not forced into copyleft. the companion software papers describe the public apis and reproducible examples; the present paper concentrates on the statistical foundations.

three contributions are advanced. first, a unified notation for the five canadian sources is established (section 1) that permits a single estimator family to operate across all of them. second, a ten-estimator per-individual ensemble is derived in its mrnj instantiation (section 3) together with a parallel aggregate  $\chi^2$  family (section 4) and a mandela rules classifier (section 5) that operates at both the federal and provincial level. the aggregate family is exercised empirically to reproduce every published  $\chi^2$  statistic from the jprott--ðoob--iftene record to within 0.01 (jprott and ðoob 1993, iftene and ðoob 1994). third, the framework is extended into the spatio-temporal domain through a self-exciting fiawkes process specification (section 6) applied to thoronto police service open data.

a headline empirical finding is reported in section 6: the provincial ontario mandela “torture”-classified proportion rises from 12.5% in fiscal 1993 to 20.6% in fiscal 1998, exceeding by 10.7 percentage points at peak the federal juv rate of 9.9% that jprott and ðoob (1993) report for fiscal 1989--1991. under the broader restrictive-confinement classification advanced in section 6, the 1998 rate reaches 40.9%.

the rest of the paper is organised as follows. section 1 establishes the notation and the five source schemas. section 3 derives the ten-estimator per-individual ensemble. section 4 presents the  $\chi^2$  family for aggregate contingency tables. section 5 introduces the mandela rules classifier and reports the empirical headline. section 6 maps the deterrence / routine-activities / certainty mechanism categorisation onto the estimator family. section 7 specifies the fiawkes self-exciting process for 0/f data. section 8 summarises the implementation in mrjie. section 9 concludes.

## A. notation and the five canadian sources

### A.i. notation

Let  $i \in \{1, \dots, n\}$  index individuals (in  $\mathcal{O}I$ ) or events (in  $\mathcal{O}E$ ), and let  $t \in \{t_0, \dots, t_T\}$  index discrete time points (fiscal quarters for  $\mathcal{O}I$ , calendar days for  $\mathcal{O}E$ , fiscal years for the aggregate sources). Each unit  $i$  at time  $t$  carries covariates  $X_{it} \in \mathbb{R}^p$ , a binary treatment indicator  $D_{it} \in \{0, 1\}$ , and an outcome  $Y_{it} \in \mathbb{R}$ . The treatment indicator denotes the presence of the analytical treatment under study, such as a mandela-classified placement, a  $\mathcal{F}$  section  $\mathcal{Z}$  trigger, or a  $\mathcal{O}E$  major-crime category.

For aggregate sources, let  $a \in \{1, \dots, A\}$  index categories of a contingency table and let  $j \in \{1, \dots, J\}$  index a stratifying variable (e.g. jurisdiction, fiscal year, demographic group). The observed count is  $N_{aj}$  and the expected count under independence is  $\hat{E}_{aj}$ .

Causal estimands are defined under the standard potential-outcomes framework:  $Y_{it}(d)$  denotes the outcome the unit would have realised had its treatment been set to  $d \in \{0, 1\}$ . The identifying assumptions of conditional exchangeability  $\{Y(0), Y(1)\} \perp\!\!\!\perp D \mid X$ , positivity  $0 < \Pr(D = 1 \mid X) < 1$ , and the stable-unit-treatment-value assumption are maintained throughout unless explicitly weakened.

### A.A. The five sources

The five canadian sources that the framework integrates are summarised in [table i](#).

**table i:** The five canadian carceral, police, and oversight data sources integrated by the  $\mathcal{M}I$  framework.

| source                                         | producer                    | unit                  | period                           | access        |
|------------------------------------------------|-----------------------------|-----------------------|----------------------------------|---------------|
| $\mathcal{O}I$ $\mathcal{C}I$ - $\mathcal{C}I$ | $\mathcal{M}I$ (ontario)    | individual--placement | $\mathcal{A}I$ --present         | open          |
| $\mathcal{F}$ $\mathcal{I}$                    | federal $\mathcal{I}$       | quarterly aggregate   | $\mathcal{A}I$ -- $\mathcal{A}I$ | open          |
| ontario $\mathcal{F}$                          | ontario $\mathcal{F}$       | investigation report  | $\mathcal{A}I$ --present         | open (scrape) |
| $\mathcal{O}E$                                 | toronto Police              | event--day            | $\mathcal{A}I$ --present         | open          |
| $\mathcal{C}I$                                 | Public Safety $\mathcal{C}$ | annual aggregate      | $\mathcal{A}I$ --present         | open          |

$\mathcal{O}I$   $\mathcal{C}I$ - $\mathcal{C}I$  is the ontario ministry of community safety and correctional services restrictive-confinement detailed dataset, a person-day-level record that includes age band, gender, alert flags, segregation type, and exit code. It is the only source in the table that supports per-individual estimands directly. The schema is published with each annual release.

$\mathcal{F}$   $\mathcal{I}$  comprises the four quarterly reports of the federal structured intervention unit implementation advisory panel ( $\mathcal{A}I$ -- $\mathcal{A}I$ ) and the parallel  $\mathcal{P}$ rott-- $\mathcal{D}$ oob-- $\mathcal{I}$ ftene academic replication corpus ( $\mathcal{P}$ rott and  $\mathcal{D}$ oob  $\mathcal{A}I$ ,  $\mathcal{I}$ ftene and  $\mathcal{D}$ oob  $\mathcal{A}I$ ), which includes contingency tables that report the federal mandela rate of 9.9% used as the reference point in [section v](#).

ontario  $\mathcal{F}$  comprises the case-disposition reports published by the ontario special investigations unit at [siu.on.ca](http://siu.on.ca). The framework's data pipeline ingests these reports through a maintained scraper shipped with  $\mathcal{M}I$  ([ruhela  \$\mathcal{A}I\$ ,b](#)). The corpus covers the  $\mathcal{F}$ 's operational history.

$\mathcal{O}E$  open data is the toronto Police service's daily event-level release of crime occurrences with

spatial coordinates aggregated to neighbourhood, supporting the spatio-temporal fiawkes specification of section 3.

ccrfj, the corrections and conditional release statistical overview, is the principal federal aggregate release. it enters the record through its tabling under the federal court of canada docket 0-239-14 (the affidavit-based admissibility process referred to in 300b 1413) and provides the federal denominator for several of the framework’s aggregate  $\chi^2$  statistics.

### 1.3. time-alignment and the panel structure

the five sources do not share a fiscal calendar. 2017 uses the ontario fiscal year (april--march), 170 14? the federal fiscal year, 017 the calendar year, and ccrfj the federal fiscal year. the framework reconciles these through a canonical fiscal quarter index implemented as an interval map in mrml’s align\_fiscal() routine (rubela 141413).

## 3. Per-individual estimators: the mrml ensemble

the per-individual ensemble specifies ten estimators of the average treatment effect (a0e) and the average treatment effect on the treated (a00) under the identifying assumptions stated in section 1. the estimators are grouped into five families.

### 3.1. inverse-probability-weighting estimators

the first family is built around the propensity score  $e(X) = \Pr(D = 1 \mid X)$ . the fiorvitz--thompson r7u estimator of the a0e is

$$\hat{\tau}_{\text{IPW}} = \frac{1}{n} \sum_{i=1}^n \left[ \frac{D_i Y_i}{\hat{e}(X_i)} - \frac{(1 - D_i) Y_i}{1 - \hat{e}(X_i)} \right], \quad (\text{i})$$

and the stabilised r7u (sr7u) variant is

$$\hat{\tau}_{\text{sIPW}} = \frac{\sum_i D_i Y_i / \hat{e}(X_i)}{\sum_i D_i / \hat{e}(X_i)} - \frac{\sum_i (1 - D_i) Y_i / (1 - \hat{e}(X_i))}{\sum_i (1 - D_i) / (1 - \hat{e}(X_i))}. \quad (\text{1})$$

the framework follows cole and fernán (141413) and trims extreme weights at the [0.01, 0.99] quantiles of  $\hat{e}$  to reduce variance inflation at the cost of a small finite-sample bias.

the a00 analogue,

$$\hat{\tau}_{\text{IPW}}^{\text{ATT}} = \frac{\sum_{D_i=1} Y_i}{\sum_{D_i=1} 1} - \frac{\sum_{D_i=0} \hat{e}(X_i) / (1 - \hat{e}(X_i)) Y_i}{\sum_{D_i=0} \hat{e}(X_i) / (1 - \hat{e}(X_i))}, \quad (\text{3})$$

follows mbens and rubin (141413).

### 3.1. augmented inverse-probability weighting

the doubly-robust a7u estimator combines the propensity score with a working outcome regression model  $\mu_d(X) = \mathbb{E}[Y \mid X, D = d]$ . the estimator is

$$\hat{\tau}_{\text{AIPW}} = \frac{1}{n} \sum_{i=1}^n \left[ \hat{\mu}_1(X_i) - \hat{\mu}_0(X_i) + \frac{D_i(Y_i - \hat{\mu}_1(X_i))}{\hat{e}(X_i)} - \frac{(1 - D_i)(Y_i - \hat{\mu}_0(X_i))}{1 - \hat{e}(X_i)} \right]. \quad (\text{4})$$

airu is consistent when either  $e$  or  $\mu_d$  (but not necessarily both) is consistently estimated; this is the double-robustness property emphasised in [robins et al. \(1994\)](#) and [bang and robins \(1998\)](#). The framework follows the implementation conventions of [funk et al. \(1999\)](#).

### 3.3. Double-machine-learning interactive regression

The interactive regression model (IRM) of [chernozhukov et al. \(1998\)](#) relaxes the parametric assumptions on  $\mu_d$  by allowing it to be estimated with arbitrary machine-learning methods, provided that the resulting nuisance estimators converge at sufficient rates for the Neyman-orthogonal score function to remain locally insensitive to nuisance error. Concretely, for the partial-linear regression (PLR) model  $Y = \theta D + \ell(X) + U$  with  $D = m(X) + V$  and  $\mathbb{E}[U \mid X, D] = \mathbb{E}[V \mid X] = 0$ , the Neyman-orthogonal score is

$$\psi_{\text{PLR}}(W; \theta, (\ell, m)) = (Y - \ell(X) - \theta(D - m(X)))(D - m(X)), \quad (6)$$

and the score has the property  $\partial_\eta \mathbb{E}[\psi_{\text{PLR}}] = 0$  at the true  $\eta = (\ell, m)$ . For the interactive regression model (IRM) with  $D \in \{0, 1\}$ ,  $\mathbb{E}[Y \mid D = d, X] = g_d(X)$  and  $\mathbb{E}[D \mid X] = m(X)$ , the airu-style orthogonal score is

$$\psi_{\text{IRM}}(W; \theta, (g_0, g_1, m)) = g_1(X) - g_0(X) + \frac{D(Y - g_1(X))}{m(X)} - \frac{(1 - D)(Y - g_0(X))}{1 - m(X)} - \theta, \quad (7)$$

with the corresponding orthogonality  $\partial_\eta \mathbb{E}[\psi_{\text{IRM}}] = 0$  at  $\eta = (g_0, g_1, m)$  ([chernozhukov et al. \(1998\)](#)). The `irm` instantiation wraps the `r` package `doubleml` ([bach et al. \(1999\)](#)) via its `doubleml_irm` class with the `mlr3` learner ecosystem ([lang et al. \(1999\)](#)), and the Python package `doubleml` via its `doubleml_irm` class with the scikit-learn learner interface ([bach et al. \(1999\)](#), [Pedregosa et al. \(1999\)](#)). The `irm` contributions are (i) the choice of sample-splitting fold count for `cvlf` (five folds, two repetitions) calibrated to the panel dimensions of `airu-rdd`, and (ii) a wrapper convention that exposes both `airu` and `airu` in a single call.

### 3.4. Propensity-score matching

Matching estimators construct a counter-factual sample by pairing each treated unit with one or more untreated units whose propensity score is within a caliper of the treated unit's. The framework implements one-to- $k$  caliper matching with replacement following [J Stuart \(1999\)](#), with default  $k = 4$  and caliper width  $0.2 \times \text{SD}(\logit e)$  per [Austin \(1999\)](#). The matched-sample `airu` is

$$\hat{\tau}_{\text{match}}^{\text{ATT}} = \frac{1}{n_1} \sum_{i: D_i=1} \left[ Y_i - \frac{1}{k} \sum_{j \in \mathcal{M}(i)} Y_j \right], \quad (8)$$

where  $\mathcal{M}(i)$  denotes the matched-control set for treated unit  $i$ .

### 3.5. Partial-linear and super-learner variants

The partial-linear regression model (PLR) of [chernozhukov et al. \(1998\)](#) specifies  $Y_i = D_i\theta_0 + g_0(X_i) + U_i$ , with  $\mathbb{E}[U_i \mid X_i, D_i] = 0$ , and recovers  $\theta_0$  through orthogonalisation against  $\hat{g}_0$ . The super-learner variant of [van der Laan et al. \(1999\)](#) replaces the single nuisance estimator with a cross-validated convex combination of candidate learners. The `irm` instantiation uses `glmnet` ([Friedman et al. \(1999\)](#)) and `ranger` ([Wright and Ziegler \(1999\)](#)) as the candidate library for both

$\hat{g}_0$  and  $\hat{m}_0$ , with weights chosen by ten-fold cross-validation on the in-sample squared-error loss.

### 3.D. Sensitivity analysis

each estimand is reported with an  $\varepsilon$ -value (vanderweele and ding 2013), the minimum unmeasured-confounder--outcome and unmeasured-confounder--treatment association required to explain away the observed effect under the worst-case unmeasured-confounder distribution. The  $\varepsilon$ -value is defined on the risk-ratio scale. For an effect with risk ratio  $\widehat{RR}$ ,

$$\text{E-value}_{RR} = \widehat{RR} + \sqrt{\widehat{RR}(\widehat{RR} - 1)}. \quad (\Theta)$$

because odd placements are common ( $p > 0.5$  in some cohorts), the rare-outcome odds-ratio  $\rightarrow$  risk-ratio approximation breaks down. We therefore report both the rr-scale  $\varepsilon$ -value (eq.  $\Theta$ ) using the zhang--yu approximation (zhang and yu 1999) and the corresponding odds-ratio  $\varepsilon$ -value

$$\text{E-value}_{OR} = \widehat{OR} + \sqrt{\widehat{OR}(\widehat{OR} - 1)}, \quad (\Theta)$$

and treat the spread between eqs. ( $\Theta$ ) and ( $\Theta$ ) as a sensitivity range. values exceeding 2.0 are conventionally interpreted as robust to plausible unmeasured confounding by clinical-severity covariates that are not available in the open amb-redd record.

### 3.X. generalised-estimating-equation clustering

when the unit of analysis is the placement-record (a single individual may contribute many records over a fiscal year) the within-person correlation between observations must be absorbed. The mrnj clustering grid extends the per-individual estimators with a generalised-estimating-equation (gee) clustering step over four candidate cluster keys: institution, regional cluster, individual, and a combined institution--quarter key. each candidate yields a sandwich-variance estimate; the framework reports all four to allow sensitivity-to-clustering inspection (fubbard et al. 2018; cameron and miller 2018).

## 4. aggregate estimators: the $\chi^2$ family

for aggregate sources (jur id?, ccrf?) the unit of observation is a count, not an individual. The mrnj aggregate family specifies  $\chi^2$  statistics under various marginal-fixing conventions to recover the contingency-table comparisons reported in the published fprott--doob--iftene record.

for an  $A \times J$  table with observed cell count  $N_{aj}$ , total  $N_{..}$ , row total  $N_{a.}$ , and column total  $N_{.j}$ , the standard  $\chi^2$  statistic under the independence null is

$$\chi^2 = \sum_{a=1}^A \sum_{j=1}^J \frac{(N_{aj} - \hat{E}_{aj})^2}{\hat{E}_{aj}}, \quad \hat{E}_{aj} = \frac{N_{a.}N_{.j}}{N_{..}}. \quad (\text{iv})$$

The framework's  $\chi^2$  family includes (i) the standard statistic, (ii) the Yates-continuity-corrected statistic for  $2 \times 2$  subtables, (iii) the Cochran--Armitage trend test for ordered  $A$ , and (iv) the Mantel--Haenszel statistic for stratified tables. empirical

replication of every published fprott--dooob--iftene  $\chi^2$  to within 0.01 is demonstrated in the mcrre test suite (ruhela 2023).

## e. mandela rules classifier and the headline finding

The united nations standard minimum rules for the treatment of prisoners (the mandela rules) define solitary confinement under two coupled conditions (united nations general assembly 1955): rule 43 names a placement prolonged when a single continuous placement exceeds fifteen days; rule 44 names a placement solitary when the individual is held twenty-two or more hours per day without meaningful human contact. Both conditions must hold for a placement to be mandela-torture under any a/ref/yh/iv. The mcrre mandela rules classifier operationalises both conditions where the data supports them, and defaults to a duration-only proxy with explicit caveats where the meaningful-contact field is unavailable (see section 2.3).

### e.i. federal operationalisation

at the federal level, the fprott--dooob--iftene record (fprott and doob 2013) classifies a section 34 structured intervention unit placement as mandela-torture when the recorded placement duration exceeds fifteen days and one or more of the “meaningful contact” criteria is unmet. Their published rate for fiscal 2019–2020 is 9.9% of all section 34 placements. The mcrre implementation replicates this classification logic on the same data and recovers 9.9% to within rounding.

### e.2. provincial operationalisation

at the provincial level, othf aui-rd does not include the direct “meaningful contact” fields available in the federal data. The framework therefore operationalises the mandela classification on duration alone, applied to the segregation-type placements recorded in aui-rd. The conservative rate uses the strict >15-day duration threshold; the broader restrictive-confinement rate uses the union of segregation-type placements and high-alert placements satisfying the same duration threshold.

### e.3. cross-jurisdiction comparison: an apples-to-apples spectrum, not a single number

Direct comparison of the federal 9.9% (fprott and doob 2013) with a single provincial rate is methodologically delicate for two reasons: (i) the federal rate is computed under both rule 43 (duration) and rule 44 (meaningful contact), whereas othf public release does not expose meaningful-contact fields; (ii) federal numbers are at the person-stay level, provincial at the ci individual-cumulative level within fiscal year. We therefore report a spectrum of estimates rather than a single headline (table 1).

The substantive direction is robust: the provincial duration-only estimate (A) exceeds the federal strict estimate by +10.7 percentage points at the 2020 peak, and the broader-rd estimate is larger still (40.9%). The magnitude of the gap, however, is definitional. A like-for-like provincial estimate under rule 43 and rule 44 would land between the duration-only ceiling (A) and the no-alert proxy floor (C). Pending a ci-level linkage to bui alert flags (queued for the next release of mcrre\_classify\_mandela()), we report this spectrum rather than a single number. On the federal side, the fprott--dooob 2013 single headline of 9.9%



Table A: mandela rules spectrum.  $A$  (rule  $q_3$  only, *any individual-cumulative*) is the upper-bound provincial rate.  $B_{\text{any-alert}}$  and  $C_{\text{no-alert}}$  are *any placement-level sensitivities* using the *any alert* columns as a proxy for staff contact (any-alert presence implies medical / suicide-watch staff visits and is therefore the looser bound on rule  $q_4$ ; the no-alert subset is the strictest contact-reduction proxy). Federal rate is *prott--doob* *any*’s strict duration-plus-meaningful-contact figure.

| jurisdiction (period)                       | rule $q_3$ only<br>( $A$ ) | rule $q_3 \cap$<br>alerted ( $B$ ) | rule $q_3 \cap$<br>no-alert ( $C$ ) | rc<br>broader               | source                         |
|---------------------------------------------|----------------------------|------------------------------------|-------------------------------------|-----------------------------|--------------------------------|
| federal <i>any</i> <i>any</i> -- <i>any</i> | $\geq 10\%$ <sup>a</sup>   | ---                                | ---                                 | $\approx 28\%$ <sup>b</sup> | 9.9% <i>prott and doob any</i> |
| provincial ontario <i>any</i>               | 12.5%                      | 0.072%                             | 0.003%                              | 31.5%                       | mrm                            |
| provincial ontario <i>any</i>               | 16.5%                      | 0.463%                             | 0.194%                              | 36.0%                       | mrm                            |
| provincial ontario <i>any</i>               | 20.6%                      | 0.905%                             | 0.387%                              | 40.9%                       | mrm                            |

<sup>a</sup> *prott--doob--iftene any* reports 10% of  $N = 1,979$  *any* person-stays met mandela “torture” (rule  $q_3 \cap$  rule  $q_4$ , prolonged  $\geq 15$  days  $\cap \geq 22$  hr no meaningful contact); rule  $q_3$  alone (duration  $\geq 15$  days, without imposing the contact criterion) is a strict superset and is therefore bounded below by this figure. an exact federal duration-only sub-rate is not separately published in the *prott--doob any*/*any* series.

<sup>b</sup> *prott--doob--iftene any* reports 28% of the same cohort met mandela “solitary confinement” (rule  $q_4$ , no contact criterion, any duration), used here as the most defensible federal analogue of the broader-rc column.

can now be bracketed against the underlying *prott--doob--iftene any* cohort ( $N = 1,979$ ): a federal duration-only sub-rate ( $A_{\text{fed}}$ ) is bounded below by 10% (the published torture rate is a subset of rule  $q_3$ ), and the federal contact-only sub-rate is 28%, so the federal rule  $q_3 \cap$  rule  $q_4$  cell is the intersection of these and the federal rule  $q_3$  ceiling lies in  $[10\%, 100\%]$ . The provincial *any* rule  $q_3$  cell (20.6%) falls within the most defensible portion of that range. The substantive implication for the jahn-settlement oversight framework is that the published duration-only metrics under-state the structural problem on the upper end and the no-alert proxy under-states it on the lower end --- both readings still place the provincial problem at or above the federal benchmark, with the substantive direction surviving any definitional refinement.

#### B. Caveats

Three caveats apply. First, the provincial classifier operates on duration alone; absent “meaningful contact” criteria, the classification is necessarily duration-conservative and may under-count placements that would qualify on additional criteria. Second, *any* *any*--*any* identifiers are reassigned at fiscal boundaries; within-person placement chaining is therefore restricted to single-fiscal-year windows. Third, the broader-restrictive-confinement rate uses a union rule (segregation-type or high-alert) that the *prott--doob--iftene* record does not separately validate; readers who require strict replication should restrict attention to the mandela-classified row.

## D. Mechanism Categorisation

Each estimator in the per-individual ensemble of Section 3 can be tied to one or more substantive sociolegal mechanisms following the categorisation of *zorro medina* (*any*) and its precursors in *becker* (*any*), *cohen and felson* (*any*), and *apel* (*any*). The framework distinguishes three mechanism classes.



deterrence. The deterrence mechanism applies when the treatment changes the expected utility of an action by altering its perceived cost. Estimands under deterrence are typically interpreted as the effect of a sentencing or oversight regime on the rate of subsequent behaviour. The IPTW and AIPW estimators of section 3.1 and section 3.4 provide the natural decomposition:  $\hat{\tau}_{\text{AIPW}}$  separates the regression component (the deterrent effect transmitted through expected outcomes) from the weighting component (the deterrent effect transmitted through selection-into-treatment).

routine activities. The routine-activities mechanism applies when the treatment changes the convergence in space and time of motivated offenders, suitable targets, and capable guardians (Cohen and Felson 1979). Estimands under this mechanism are appropriately specified at the neighbourhood-day level (the Hawkes specification of section 4) rather than the individual-placement level.

certainty. The certainty mechanism applies when the treatment changes the perceived probability of detection or sanction rather than its magnitude. The propensity-score matching and partial-linear estimators of section 3 provide the estimand: by matching on the covariates that drive selection into treatment, the matched comparison estimates the effect under a common conditional probability of treatment given covariates.

## 4. Hawkes self-exciting process for event sequences

The Hawkes process (Hawkes 1971; Ogata 1981) provides a parsimonious generative model for event sequences with temporal clustering. The conditional intensity of a one-dimensional Hawkes process is

$$\lambda(t \mid \mathcal{H}_t) = \mu + \sum_{t_i < t} g(t - t_i), \quad (\text{ii})$$

where  $\mu > 0$  is the baseline intensity,  $\mathcal{H}_t$  is the history of events up to time  $t$ , and  $g(\cdot) \geq 0$  is the triggering kernel. The branching ratio  $n = \int_0^\infty g(s) ds$  satisfies  $n < 1$  for a stationary process.

### 4.1. Markovian (exponential) kernel

The classical Markovian Hawkes specification fixes  $g(s) = \alpha\beta e^{-\beta s}$ , yielding the conditional intensity

$$\lambda(t) = \mu + \alpha \sum_{t_i < t} \beta e^{-\beta(t-t_i)}, \quad n = \alpha, \quad (\text{i1})$$

which permits a recursive computation of the intensity through the augmented state  $S(t) = \sum_{t_i < t} \beta e^{-\beta(t-t_i)}$  satisfying  $dS/dt = -\beta S$  between events with a unit jump at each event time (Ogata 1981; Daley and Vere-Jones 1983).

### 4.2. Non-Markovian (power-law) kernel

For crime-event data the empirical autocorrelation function decays more slowly than exponentially and a power-law triggering kernel is appropriate (Möhler et al. 2011):

$$g(s) = \alpha(s + c)^{-(1+\omega)}, \quad \omega > 0. \quad (\text{i3})$$

The Hawkes specification fits both kernels by maximum likelihood and selects between them using the Bayes information criterion.

### 3.3. Spatio-temporal extension

For  $\Theta$  events with spatial coordinates  $\mathbf{x}_i \in \mathbb{R}^2$ , the spatio-temporal Hawkes intensity is

$$\lambda(t, \mathbf{x} | \mathcal{H}_t) = \mu(\mathbf{x}) + \sum_{t_i < t} g(t - t_i) f(\mathbf{x} - \mathbf{x}_i), \quad (14)$$

where  $f$  is a spatial-displacement density (typically a Gaussian or a Student- $t$ ). applied to  $\Theta$  open data, the spatio-temporal specification recovers daily-neighbourhood clustering structure that descriptive analyses do not.

### 3.4. implementation

The Hawkes implementation in `morie` provides both Markovian and power-law fits via maximum likelihood, the Ogata thinning algorithm for simulation, and a branching-ratio diagnostic for stationarity (Ruhela 2020). Goodness-of-fit is assessed through the time-rescaling theorem of Brown et al. (2001): under the fitted model the rescaled inter-event times should be exponentially distributed. A Kolmogorov-Smirnov test against the unit-rate exponential provides the goodness-of-fit  $p$ -value.

## 4. implementation in `morie`

The framework is implemented as a curated set of `morie` modules in `morie`, a dual-language (R and Python) toolkit. As of v0.3.0 the R surface is released under GPL-3.0-only and the Python surface under MIT or Apache-2.0 (rust-ecosystem dual-license). The companion software papers (Ruhela 2020, 2021) describe the public APIs and reproducible examples in detail. The present section documents only the interface between the framework’s mathematical specification and its software realisation.

The ten per-individual estimators of Section 3 are exposed as `morie::estimate_att()`, `morie::estimate_atc()`, `morie::estimate_aipw()`, and `morie::estimate_irm()` (and the corresponding Python `morie.causal.*` functions). The aggregate  $\chi^2$  family of Section 4 is exposed as `morie::chi2_doob()`. The Mandela rules classifier of Section 5 is exposed as `morie::classify_mandela()`. The Hawkes specification of Section 3 is exposed as `morie::fit_hawkes()` with explicit `kernel = 'exponential'` and `kernel = 'power_law'` options. The framework’s software DOI is <https://doi.org/10.26434/chemrxiv-2021-00000>.

## 5. conclusion

The `morie` framework brings five Canadian carceral, police, and oversight data streams together under a coordinated set of estimators and identification primitives. The ten-estimator per-individual ensemble, the aggregate  $\chi^2$  family, the Mandela rules classifier, the mechanism categorisation, and the Hawkes self-exciting specification collectively provide the statistical foundation under which empirical analyses of these sources may proceed with a single notation and a single estimator family. The framework’s empirical reach is demonstrated by the

replication of every published sprott--ðoob--iftene  $\chi^2$  to within 0.01 and by the headline finding that the provincial ontario mandela rate now exceeds the federal benchmark by 10.7 percentage points at peak.

the framework’s open-source realisation in the mörre toolkit ([ruhela 2023b](#)) makes every estimator and every mandela classification reproducible on the open ontario and federal data records. extensions of the framework to additional administrative sources (e.g. the royal canadian mounted Police annual statistics, the Public health agency of canada substance-use surveys) follow the same notational and mechanism-categorisation discipline.

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